

Oracle EPM Predictive Cash Forecasting

Trend-Based Forecasting - Advanced Lab Guide

Product Design Documentation • Implementation Guide • Training Material • Technical Reference • Troubleshooting Handbook

Advanced-stage guide

This guide explains how to design, configure, validate and use Trend-Based Forecasting in Oracle EPM Predictive Cash Forecasting. It is tailored for advanced students, consultants, treasury users and administrators who need both functional and technical depth.

Design Point	Definition
Forecast type	Trend-based methods support periodic cash forecasting only.
Use case	Line items whose future cash can be modeled from historical patterns.
Primary controls	Method selection, assumptions, historical data quality, and governance of overrides.
Audience	Cash Managers, Controllers, Admins, Architects, Oracle EPM consultants.
Business outcome	Faster, explainable forecasting for stable or pattern-driven cash lines.

BISP Trainings

1. Purpose and Learning Outcomes

Why trend-based forecasting matters

Not every cash line requires bottom-up transaction logic. Many operational and support cash lines follow recurring historical behavior. Trend methods allow the application to forecast those lines quickly and consistently while still supporting review and governance.

- Understand the role of trend-based methods in Predictive Cash Forecasting.
- Identify when trend methods are preferable to driver-based, smart-driver or manual methods.
- Configure trend methods and assumptions at the line-item and entity level.
- Run the Periodic Process Forecast rule and interpret the resulting cash forecast.
- Validate trend-based output using dashboards, comparative views and business commentary.
- Troubleshoot history gaps, wrong method selection and periodic-range design issues.

Training outcomes

Outcome	Advanced student capability
Functional design	Map each cash line to the most appropriate forecasting method.
Technical setup	Enable the feature and configure assumptions confidently.
Forecast review	Diagnose unusual results and distinguish real business change from model noise.
Governance	Document rationale for overrides and published forecast decisions.

2. Oracle Product Context

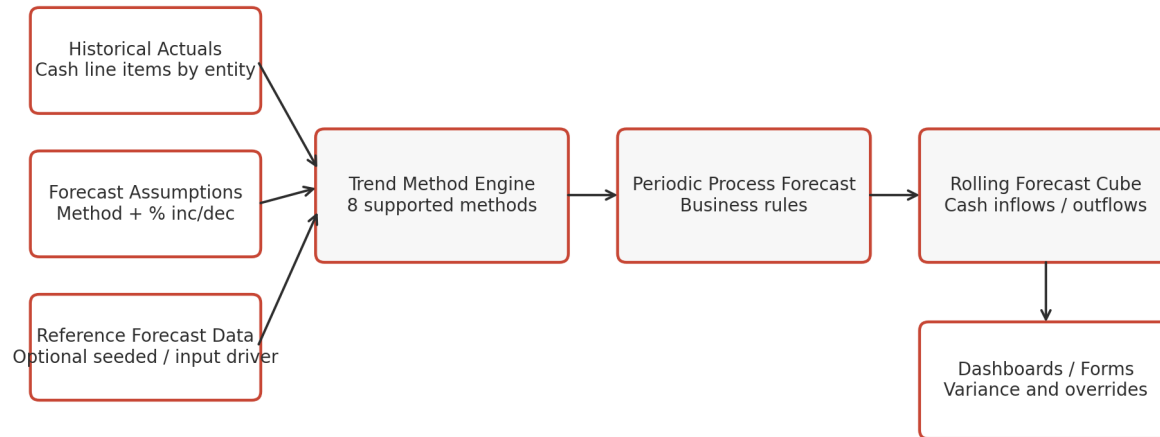
Oracle context

Oracle documents that trend-based methods can be used for cash line items where the forecast can be determined from historical trends, and that these methods are available only for periodic forecasts.

Product capability	Design implication
Trend methods are enabled during application setup	Implementation scope must include the feature early in design.
Trend methods are periodic only	Use monthly or weekly periodic horizons, not daily.
Cash Managers set trend assumptions	Ownership of assumptions should sit close to the business.
Forecast populates Rolling Forecast automatically	Validation should connect trend forms to the consumer dashboard.
Methods work best on standard patterns	Use stable, repeatable cash lines rather than volatile one-off events.

3. Trend-Based Forecasting Reference Architecture

Trend-Based Forecasting Reference Architecture



Periodic forecasts only: trend methods transform historical patterns into rolling forecast values.

The architecture combines historical actuals, trend assumptions, seeded rule logic and rolling-forecast reporting. The model is intentionally lightweight: historical data and assumptions feed the trend engine, which posts values into future periods for review and sign-off.

Recommended source candidates

Line item example	Why trend methods fit
Bank charges	Low volatility; current-year average is often sufficient.
Utilities	Latest actual often repeats with modest variation.
Travel	Prior-year average may be a useful planning anchor.
Marketing / service revenue	Prior-year period may reflect seasonal campaigns.
Trade spends	Seasonality is often visible across the year.
Rent	Year-over-year uplift can reflect contract escalation.

4. Trend-Based Methods - SmartArt Overview

Trend-Based Methods SmartArt View



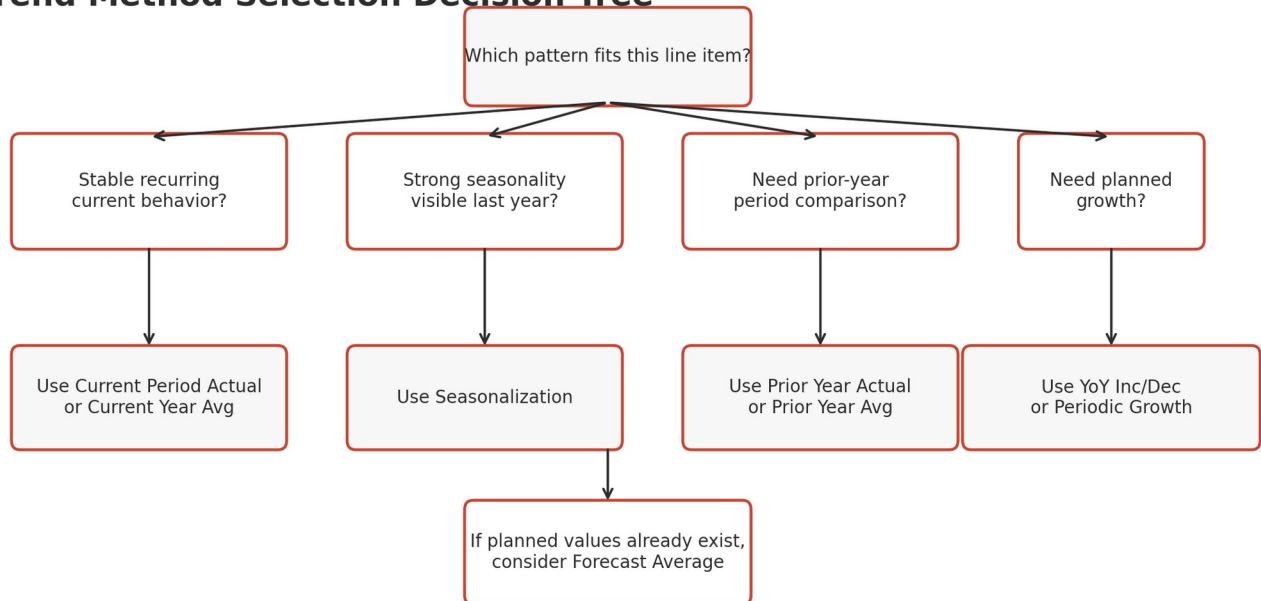
Choose methods by behavioral pattern, available history, and planning maturity.

Oracle-supported methods

Method	Logic summary	Typical example
Current Year Actual Average	Average of current fiscal-year actuals to date	Bank charges
Current Period Actual	Carry the last actual into forecast periods	Utilities
Prior Year Actual	Use the same period from prior year	Marketing or service revenue
Prior Year Actual Average	Use average of the prior fiscal year	Travel
Forecast Average	Average of current-year forecast values	Labor
Seasonalization	Apply prior-year seasonal profile to current-year average base	Trade spends
Year over Year Inc / Dec	Apply a chosen % increase or decrease to prior-year values	Rent
Periodic Growth	Calculate growth using current and prior year as the basis	Variable compensation

5. Method Selection Decision Tree

Trend Method Selection Decision Tree



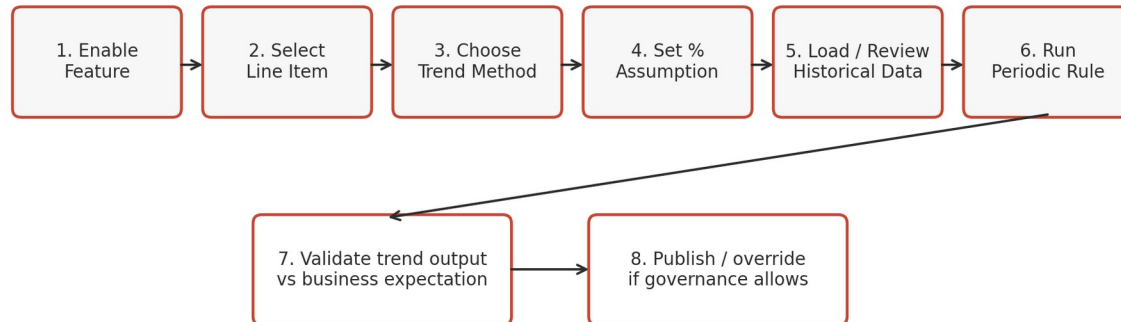
Selection principle

Choose the simplest method that captures the underlying business behavior. Simpler methods are easier to explain and govern. More complex methods should only be used when the line item truly exhibits seasonality or explicit growth dynamics.

- Stable small overheads often suit Current Year Actual Average.
- Highly recurring near-term costs often suit Current Period Actual.
- Seasonal sales-linked items often suit Prior Year Actual or Seasonalization.
- Contractual uplift items often suit Year over Year Increase / Decrease.
- Where a planning baseline already exists, Forecast Average can stabilize later periods.

6. Process Flow for Administrators and Cash Managers

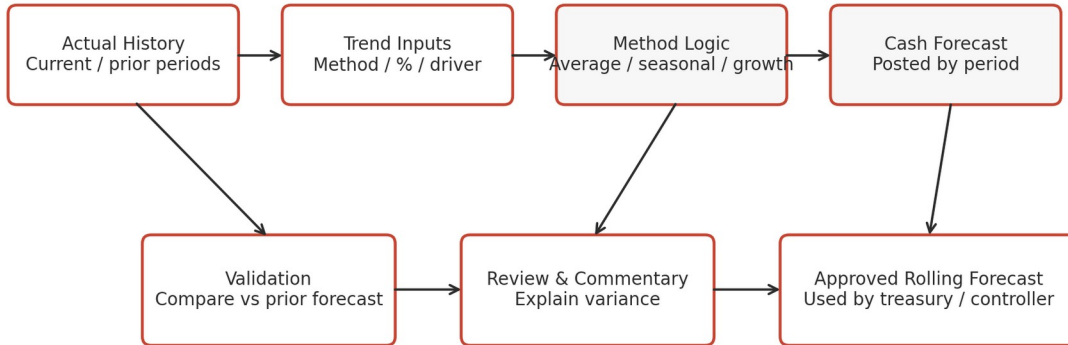
Trend-Based Forecasting Process Flow



1. Enable Trend-Based Forecasting while enabling the application or feature set.
2. Assign the trend method to the required line item / entity combinations.
3. Enter or review the trend assumption, such as a percentage increase or decrease.
4. Load or validate the historical actuals used by the method.
5. Run the Periodic Process Forecast rule.
6. Review the generated values in trend forms and Rolling Forecast forms.
7. Add commentary or perform a governed override if business intelligence suggests the pattern will break.
8. Publish the approved forecast version.

7. Data Flow and Calculation Lineage

Data Flow and Calculation Lineage



Lineage controls

Control	Why it matters	Owner
Historical completeness	Trend methods fail or mislead when prior periods are missing	Admin / Data team
Method assignment review	Wrong method selection causes systematic forecast bias	Cash Manager
Assumption sign-off	% growth assumptions should be business-approved	Controller / Treasury
Version governance	Published forecast should distinguish calculated vs overridden values	Reviewer
Variance commentary	Material changes must be explained, not merely accepted	Cash Manager

8. Oracle-Inspired Configuration Mockup

Oracle-Inspired Configuration Mockup

Home > Application > Enable Features > Trend-Based Forecasting

Forecast Range
Monthly / Weekly

Feature
Trend Enabled

Custom Dimension
Optional

Version
Base / Stress

Method Assignment
Line item + entity combination

Assumptions
% increase / decrease

Calculation Rule
Periodic Process Forecast

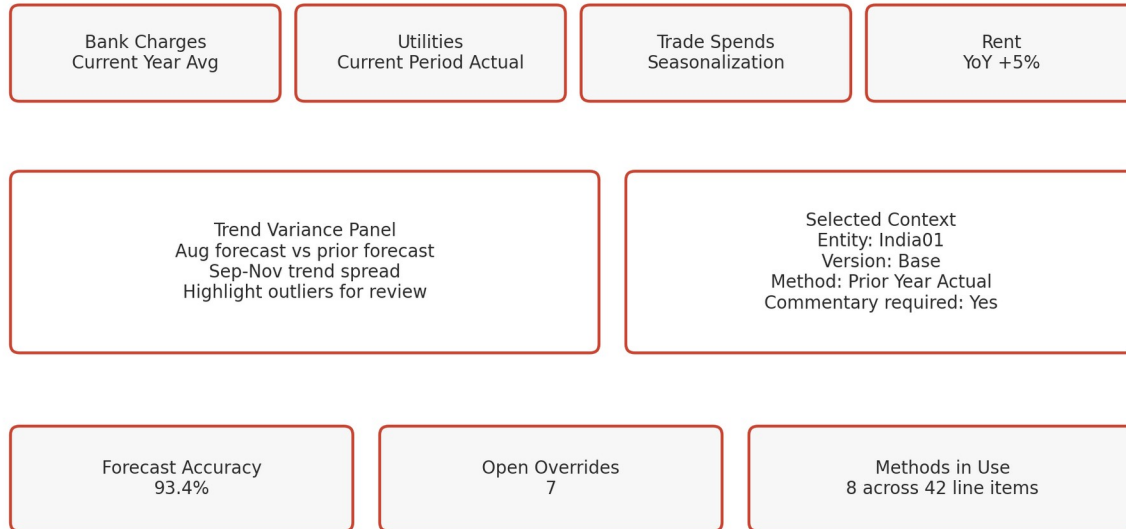
Save and Validate

Implementation checklist for configuration

- ☐ Confirm the application forecast range is periodic.
- ☐ Enable the Trend-Based Forecasting feature.
- ☐ Set up custom dimensions if additional reporting slices are needed.
- ☐ Define relevant versions and scenarios such as Base, Stress and Management Action.
- ☐ Assign methods by line item and entity combination.
- ☐ Review the assumptions input form design and user access.
- ☐ Validate that the Periodic Process Forecast rule is available and tested.

9. Dashboard and Review Experience

Trend Forecast Dashboard Mockup



- Show both the chosen trend method and the resulting amount together.
- Make outliers visually obvious to guide business review.
- Allow commentary capture where trend-based results differ materially from prior expectations.
- Display method utilization metrics so the administrator understands application behavior.
- Separate calculated values from overrides in review dashboards.

10. Advanced Lab Scenario

Scenario

A regional treasury team is preparing a 12-month monthly rolling forecast for Entity IN01. Several non-transactional line items will use trend methods: Bank Charges, Utilities, Travel, Marketing, Trade Spends, Rent, Labor and Variable Compensation.

Lab setting	Value
Entity	IN01
Horizon	Monthly periodic forecast
Version	Base
Current fiscal year actuals available through	June
Objective	Configure methods, calculate Jul-Dec forecast and validate reasonableness

11. Historical Data Used in the Lab

Line Item	Jan	Feb	Mar	Apr	May	Jun	Prior Year Avg / Ref
Bank Charges	52	49	51	50	53	49	50
Utilities	108	111	110	112	109	113	110
Travel	42	38	44	47	45	46	43
Marketing	120	95	110	145	160	130	118
Trade Spends	200	180	220	260	310	280	Seasonal
Rent	300	300	300	300	300	300	Prior Year 285
Labor	615	620	618	622	624	626	Forecast Avg
Variable Compensation	90	88	95	98	102	105	Growth driven

Amounts shown in \$000. These values are intentionally simple so students can focus on method logic rather than data-loading complexity.

12. Lab Step 1 - Assign Methods

Line Item	Chosen Trend Method	Rationale
Bank Charges	Current Year Actual Average	Stable low-volatility recurring charge
Utilities	Current Period Actual	Latest pattern is a reasonable short-term proxy
Travel	Prior Year Actual Average	Moderately recurring with small range of variation
Marketing	Prior Year Actual	Period-specific campaign timing matters
Trade Spends	Seasonalization	Strong seasonal profile expected
Rent	Year over Year Inc / Dec	Fixed recurring payment with contractual escalation
Labor	Forecast Average	Later periods should align to existing forecast baseline
Variable Compensation	Periodic Growth	Business expects growth based on current-year trend

13. Lab Step 2 - Enter Assumptions

Method / Line Item	Assumption entered
Current Year Actual Average / Bank Charges	0% change
Current Period Actual / Utilities	Use June actual as repeat basis
Prior Year Actual / Marketing	0% change in seasonal pattern
Seasonalization / Trade Spends	Apply prior-year shape to current-year average
Year over Year Inc / Dec / Rent	+5%
Periodic Growth / Variable Compensation	+4% growth across future periods
Forecast Average / Labor	Use current-year forecast average as baseline

Governance note

Only assumptions with business impact should be exposed to power users. Protect method assignment and core rule logic from casual change.

14. Lab Step 3 - Step-by-Step Calculations

Bank Charges – Current Year Actual Average: Average Jan-Jun = $(52+49+51+50+53+49) / 6 = 50.67$. Forecast Jul-Dec each ≈ 50.7 .

Utilities – Current Period Actual: June actual = 113. Forecast Jul-Dec each = 113 unless business review overrides the repetition.

Travel – Prior Year Actual Average: Prior-year average reference = 43. Forecast Jul-Dec each = 43.

Rent – YoY Increase / Decrease: Prior-year corresponding period = 285. Apply 5% increase: $285 \times 1.05 = 299.25$, rounded to 299 or 300.

Variable Compensation – Periodic Growth: If June baseline is 105 and periodic growth is 4%, Jul = 109.2, Aug = 113.6, Sep = 118.1, and so on.

Line Item	Illustrative Jul	Illustrative Aug	Illustrative Sep	Calculation logic
Bank Charges	50.7	50.7	50.7	Current-year average
Utilities	113	113	113	Current period actual
Travel	43	43	43	Prior-year average
Rent	299	299	299	Prior-year amount $\times$ 1.05
Variable Compensation	109.2	113.6	118.1	Compounding 4% growth

15. Lab Step 4 - Seasonalization Example

Trade Spends seasonalization

Oracle defines Seasonalization as applying last year's seasonality to the current year actual average. First compute the current-year average basis, then distribute future periods using the prior-year seasonal profile.

Month	Prior-Year Trade Spend	Seasonality Weight	Forecast using current-year avg basis
Jul	240	12%	288
Aug	260	13%	312
Sep	210	10.5%	252
Oct	190	9.5%	228
Nov	220	11%	264
Dec	280	14%	336

Illustrative basis assumption: the current-year average supports a remaining-period pool that is distributed according to the prior-year seasonal pattern.

16. Lab Step 5 - Forecast Output and Review

Line Item	Jul-Dec Forecast Total (\$000)	Reasonableness check
Bank Charges	304	Stable and close to historical run-rate
Utilities	678	Flat outcome; review for summer or winter utility spikes
Travel	258	Conservative baseline
Marketing	760	Depends on prior-year campaign seasonality
Trade Spends	1,680	Higher during promotional months as expected
Rent	1,794	Contractual escalation correctly applied
Labor	3,756	Aligned to forecast baseline
Variable Compensation	734	Rising with business growth assumptions

Review question

Which outputs look mechanically correct but business-risky? In this scenario, flat utilities may be mathematically valid yet operationally weak if known seasonality or rate changes are expected.

17. Trend Forecast vs Actual Review Table

Line Item	Forecast Method	Latest Forecast (Q3 Total)	Scenario Actual / Review Point	Variance	Commentary
Bank Charges	CY Actual Avg	152	149	-3	Within tolerance
Utilities	Current Period	339	360	+21	Possible tariff increase; consider override
Travel	PY Avg	129	121	-8	Conservative but acceptable
Trade Spends	Seasonalization	912	880	-32	Campaign delayed
Rent	YoY +5%	897	897	0	Matches contract

18. Product Design Guidance

- Group line items by forecasting behavior so users can review similar method types together.
- Expose the selected trend method and assumption in the same screen as the output.
- Separate trend assumptions from manual override inputs to protect model integrity.
- Use color-coded dashboard states: calculated, reviewed, overridden and approved.
- Provide drillable variance views that compare trend forecast, prior forecast and actuals.

Architecture notes

Design topic	Recommendation
Application design	Enable trend methods only if enough periodic history exists.
Dimension use	Use optional custom dimension only when it improves reporting value.
Performance	Keep unnecessary sparse intersections out of user-facing forms.
Security	Allow assumptions editing only to authorized power users.

19. Troubleshooting Handbook

Log / Error Examples

ERROR TRD-101 Trend-based methods are only supported for periodic forecasts. Review forecast range setup.

WARN TRD-118 Prior year history incomplete for selected line item. Forecast may be understated.

ERROR TRD-204 No trend method assigned for Entity IN01 / Line Item Utilities.

INFO TRD-001 Periodic Process Forecast completed successfully for Version Base.

Symptom	Likely cause	Resolution
No trend result is generated	Method not assigned or rule not run	Check method assignment and run Periodic Process Forecast
Unexpectedly flat output	Current Period Actual or average method used on a seasonal line	Reassess method selection
Seasonalization looks distorted	Prior-year history incomplete or abnormal	Validate prior-year data set and business events
Trend form unavailable	Feature not enabled	Enable Trend-Based Forecasting and refresh artifacts
Incorrect values in daily app	Trend methods used outside periodic context	Redesign using periodic forecast range
Too many overrides	Model method is not aligned to business reality	Refine the method mapping framework

20. Operations and Governance

Role	Responsibilities for trend-based forecasting
Administrator	Enable feature, manage artifacts, support rules and user access
Cash Manager	Select methods, manage assumptions, review generated forecast
Controller / Treasury Lead	Approve key assumptions and overrides
Solution Architect	Define modeling standards and design pattern library
Support Team	Monitor logs, historical loads and recurring exceptions

Governance principle

A calculated forecast should be the default starting point. Manual adjustment is allowed only when there is a documented business rationale.

21. Production Runbook

1. Load or validate periodic actual history for the required entities and line items.
2. Review method assignments and time-sensitive assumption changes.
3. Run the Periodic Process Forecast rule.
4. Check logs for completion and warnings.
5. Review dashboard variances and material exceptions.
6. Capture commentary and approve necessary overrides.
7. Publish the forecast version and archive evidence.

22. Implementation Checklist

- ☐ Define the periodic forecast horizon (monthly or weekly).
- ☐ Enable Trend-Based Forecasting.
- ☐ Catalog eligible line items by behavioral pattern.
- ☐ Document the chosen method for each eligible line item.
- ☐ Establish assumption governance and approval owners.
- ☐ Load and validate historical actual data.
- ☐ Test all eight method types with sample data.
- ☐ Design user forms, dashboards and commentary capture.
- ☐ Train cash managers and reviewers.
- ☐ Publish support model and troubleshooting steps.

23. Advanced Student Challenge

Challenge case

A student must forecast Utilities, Marketing, Insurance, Trade Spends and Variable Compensation for three legal entities. One entity has incomplete prior-year history, another has a sudden rate increase, and the third has a one-time promotional spike last year.

- Choose a suitable method for each line item and explain why.
- Identify where pure trend logic is sufficient and where manual intervention is justified.
- Propose how to handle missing history without breaking forecast governance.
- Describe the dashboard cues you would add to alert reviewers to questionable trend outputs.

24. Summary and Completion Criteria

- Trend-based forecasting is understood as a periodic, pattern-driven method within PCF.
- Users can distinguish all eight supported trend-based methods and know when to use them.
- Administrators can enable and support the feature in a production-ready design.
- Cash Managers can generate, review and explain trend-based cash forecasts.
- The organization has clear controls for assumptions, overrides and troubleshooting.
- The implementation can be used as product design documentation, implementation guide, training material, technical reference and troubleshooting handbook.

End state

A strong trend-based implementation reduces manual effort for stable cash lines while preserving business review, explainability and governance.